



Infinite Drip

A universal cryptocurrency faucet

AI agents are on-chain

176M

AI agent txs
settled on-chain
(past 12 months)

19%

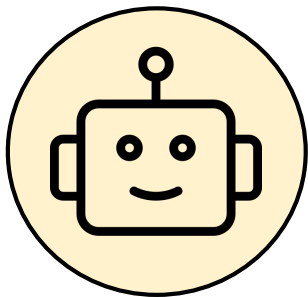
of all on-chain
txs now come
from AI agents

17,000+

AI agents live on
Web3 platforms today

Sources: Keyrock and CoinDesk, May 2025 to Apr 2026 (176M transactions); Nevermined (19% of on-chain activity, 17,000+ agents on Web3).

Problem



**Person or AI
agent**

wants to develop
& test on-chain



**needs
tokens**



Today's onramps



Exchange: needs
human



Rate limited faucets



Already hold crypto



IC3 Hackathon



In the past, anyone could join with CPU

- Now, all access to crypto is gated
 - PoS chains require funds to start
 - PoW requires ASICs
 - Testnet faucets have disgusting anti-Sybil mechanisms

0x781F4B2A9226750A51bA6Dd79c49D6A16f9ef3

The faucet denied starting a session because your IP Address is marked as hosting range.



I'm not a robot



You can verify your unique identity and increase your score using [Bitcoin Passport](#).

Ensure your provided address achieves a minimum score of 10 to initiate a session.

Passport Address:

0x87E52C2069B921420c465956E7baBE0076527B2f

Passport Score:

0 (Reward Factor: 1 ⓘ)

Start Mining

Refresh Passport

or

Verify Manually

Goals



Easy to use



Trust-minimized



No barriers to entry, **no limits**

All tokens have value.

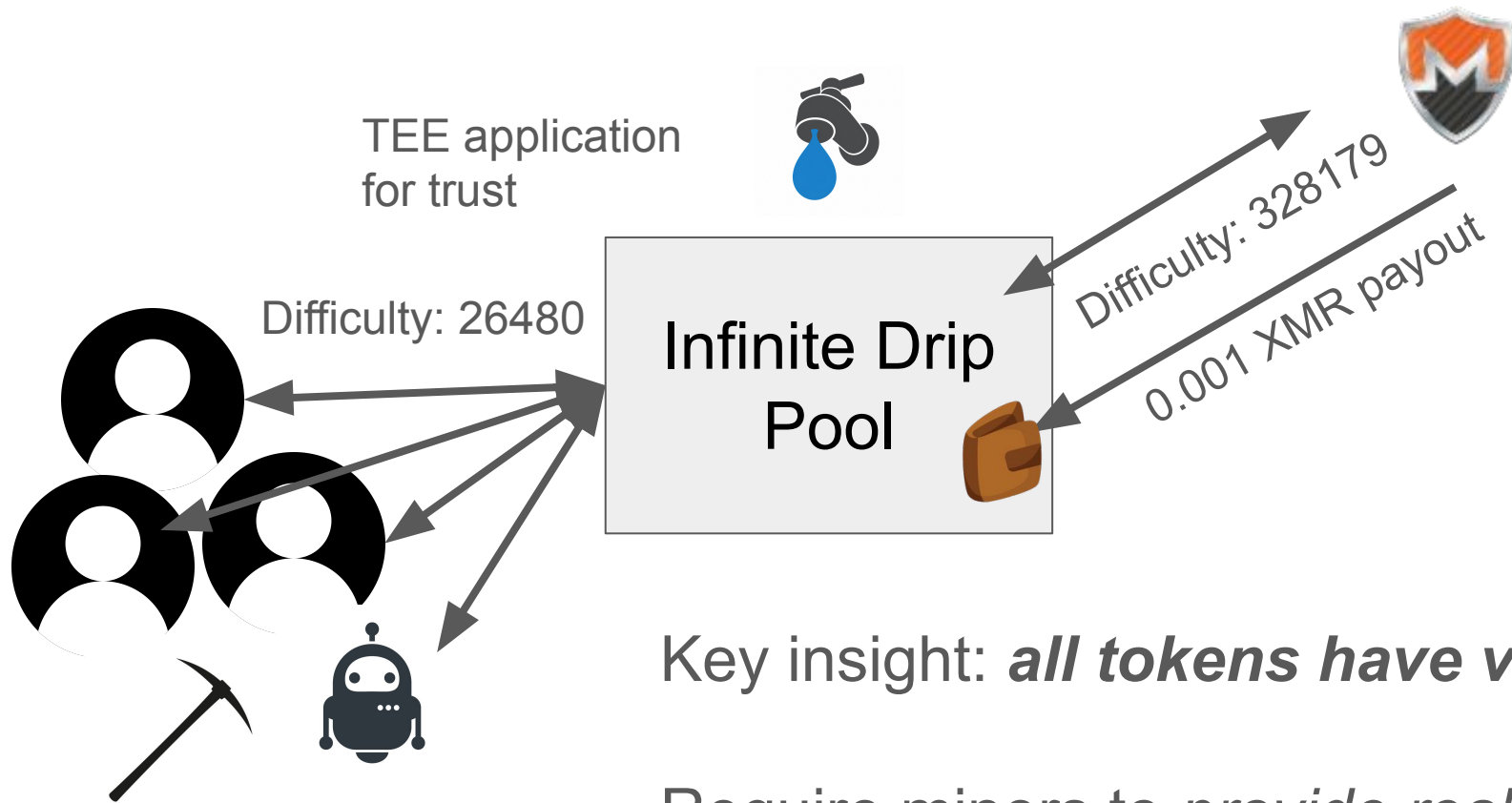
So make faucet users provide value, too!

You can mine on CPU!

- PoW chains still exist
- Some have resisted ASICs
 - RandomX: PoW algorithm **optimized for general-purpose CPUs**

Numbers

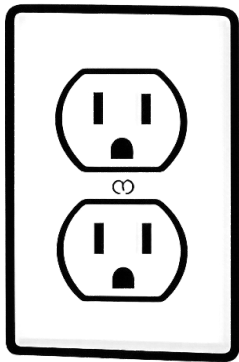
- \$0.05/day on James's laptop (2,000 H/s)
- Enough to send a transaction:
 - On Base **every 6 minutes**
 - On OP mainnet, **under 1 minute**
 - Sepolia testnet, **0.27 seconds of mining**
 - At 1 sepETH = \$0.006 (today's price)



Key insight: ***all tokens have value***

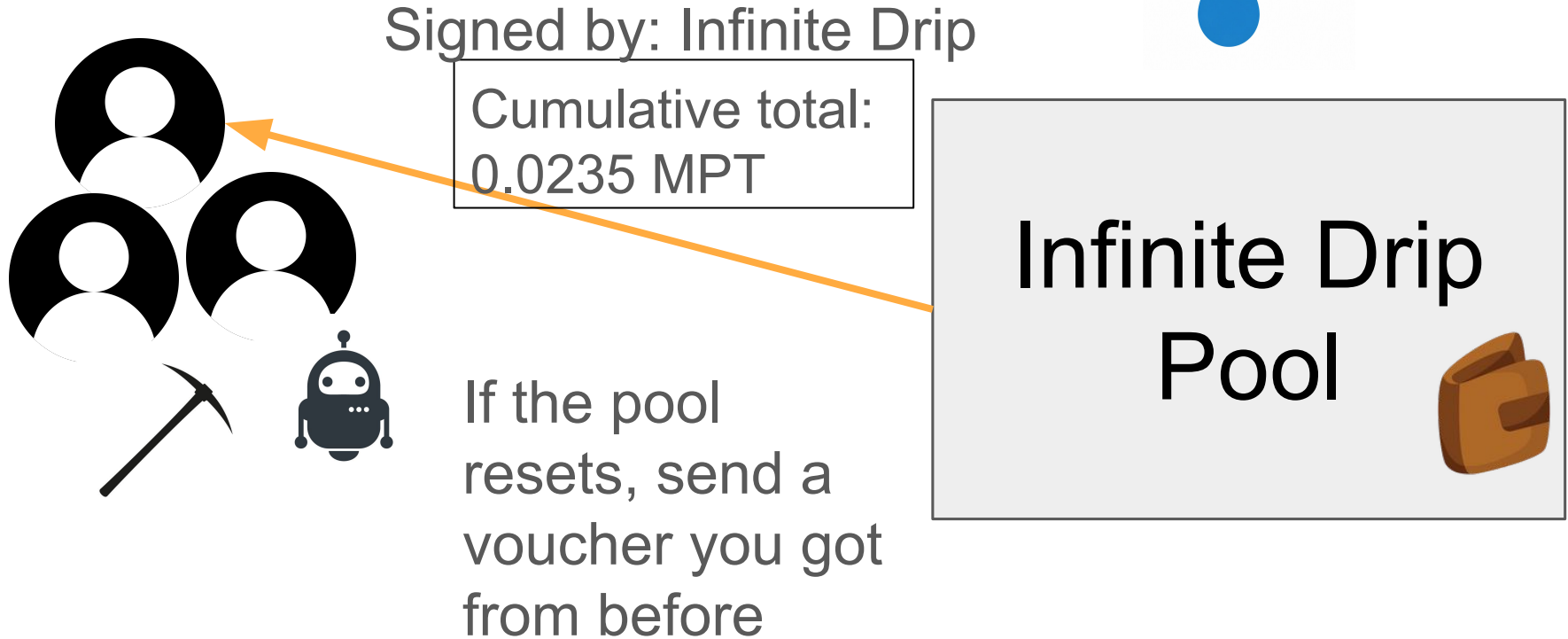
Require miners to *provide real value*

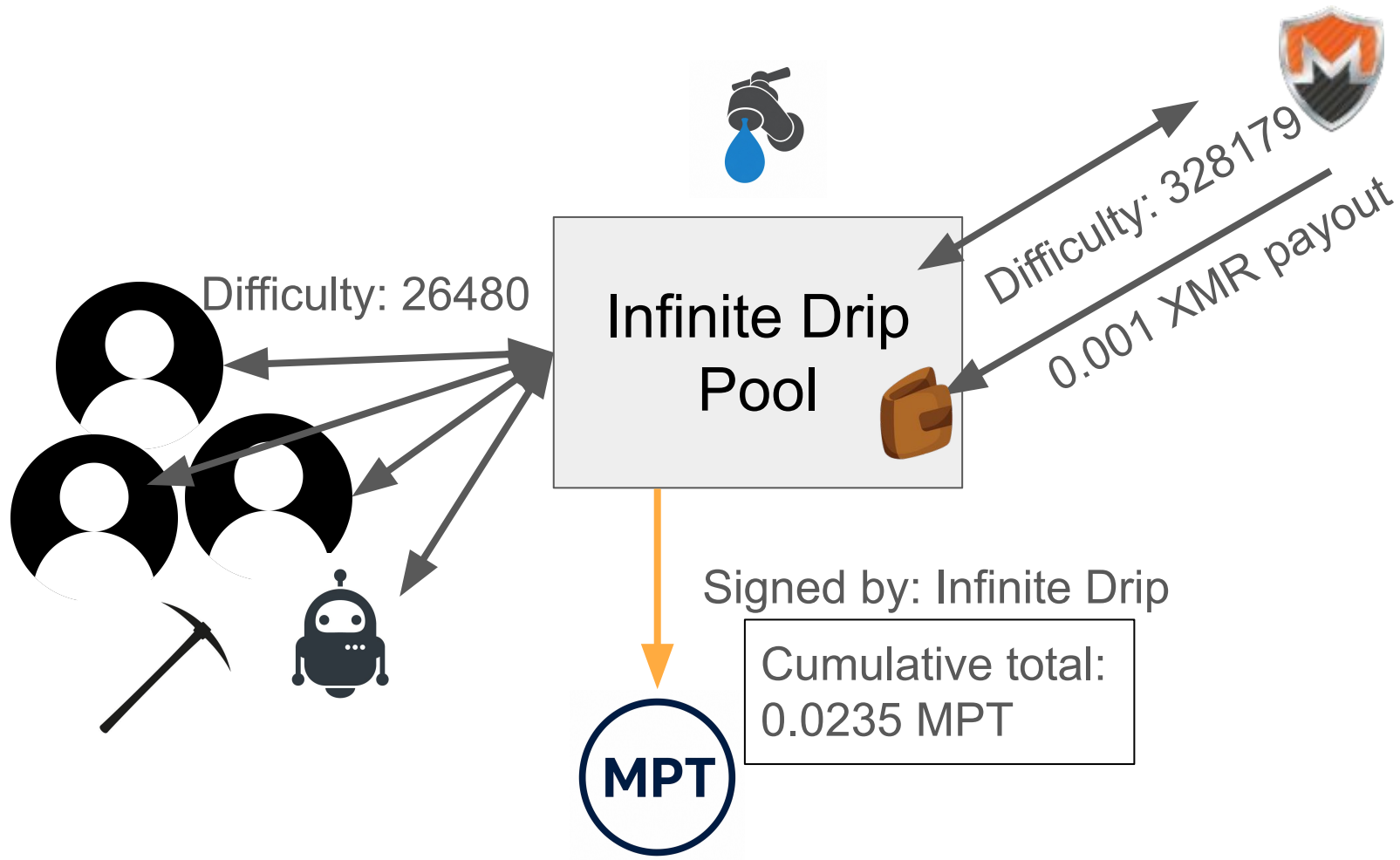
An unexpected adversary



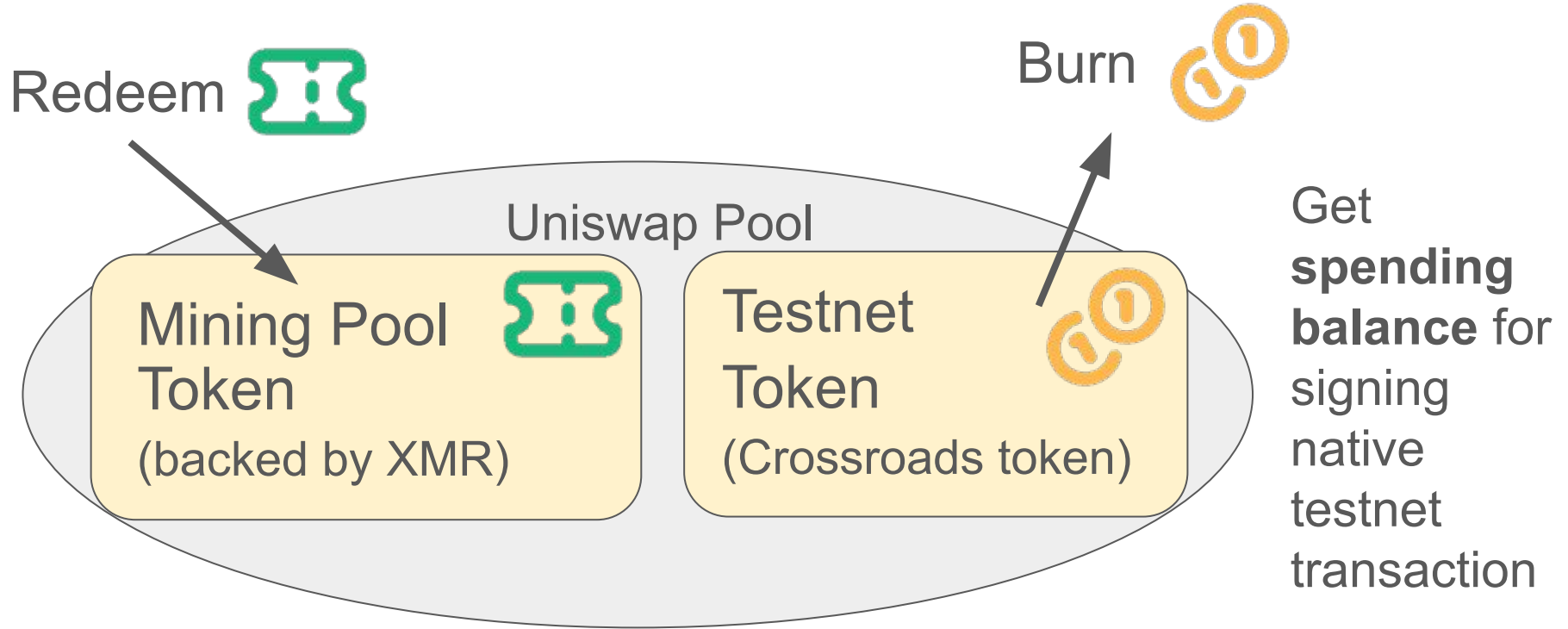
- Application container has storage...
- Only reliable backing storage is the blockchain

Payment channel to each miner



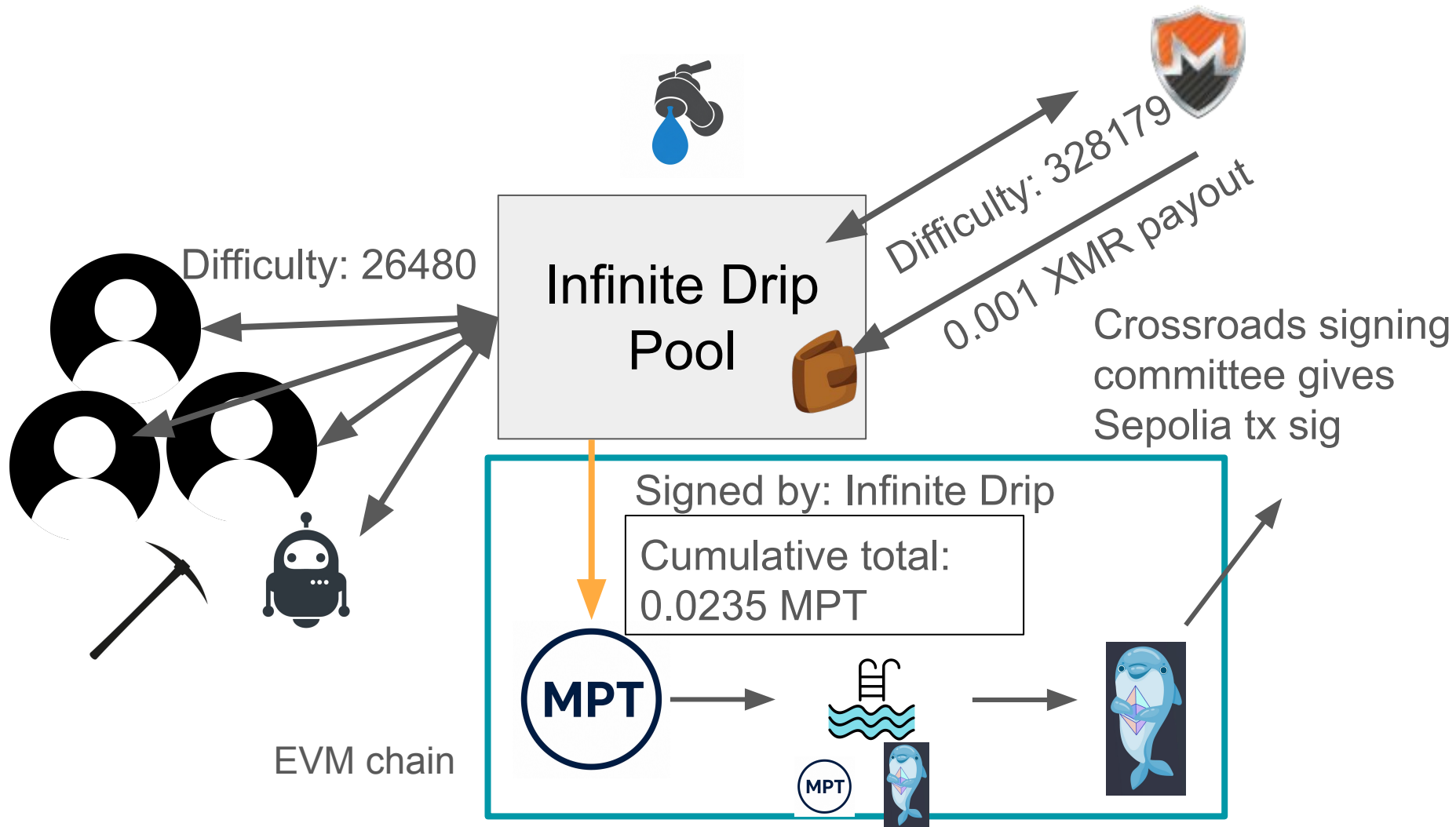


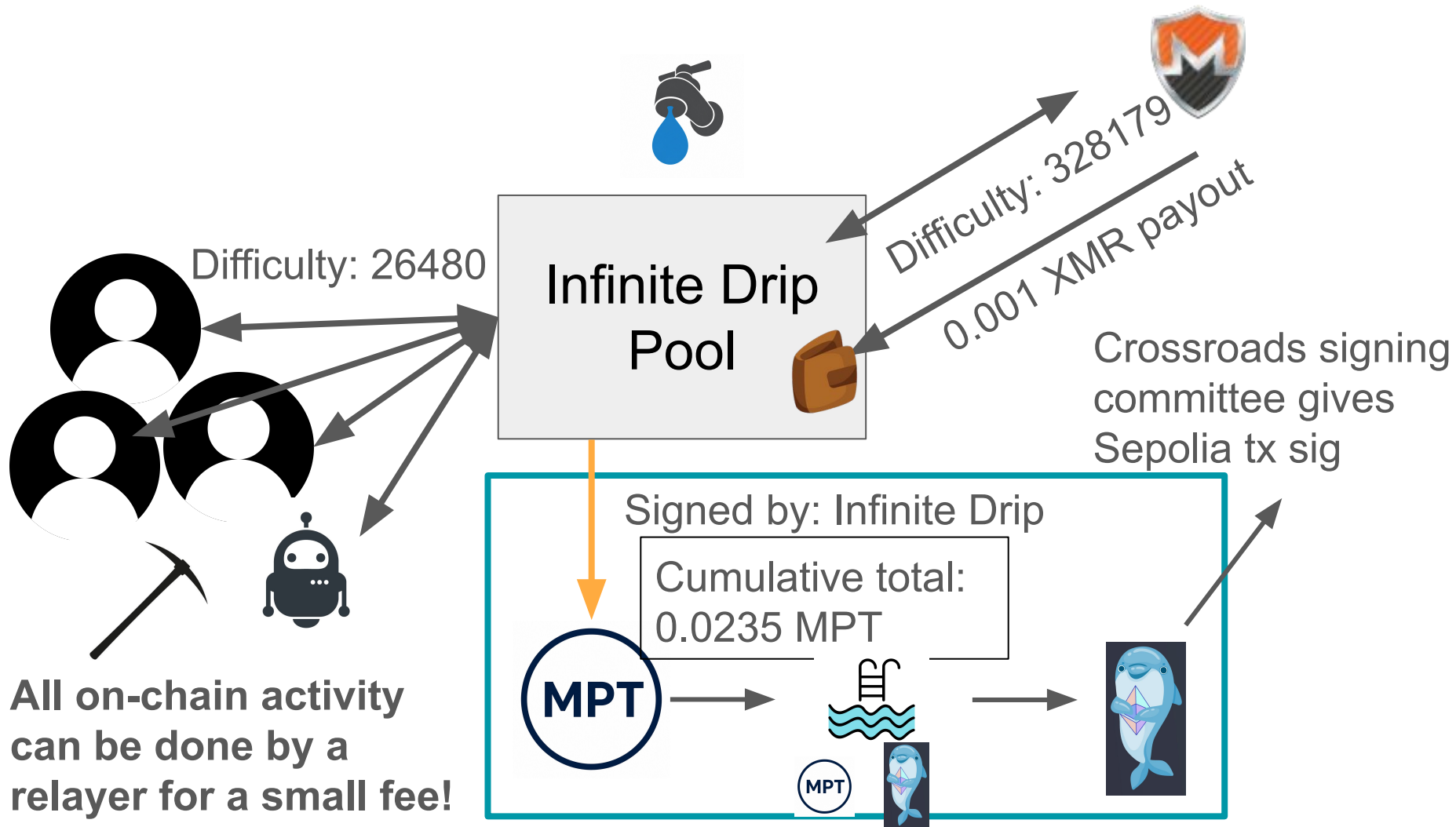
Redeeming mining credits for tokens

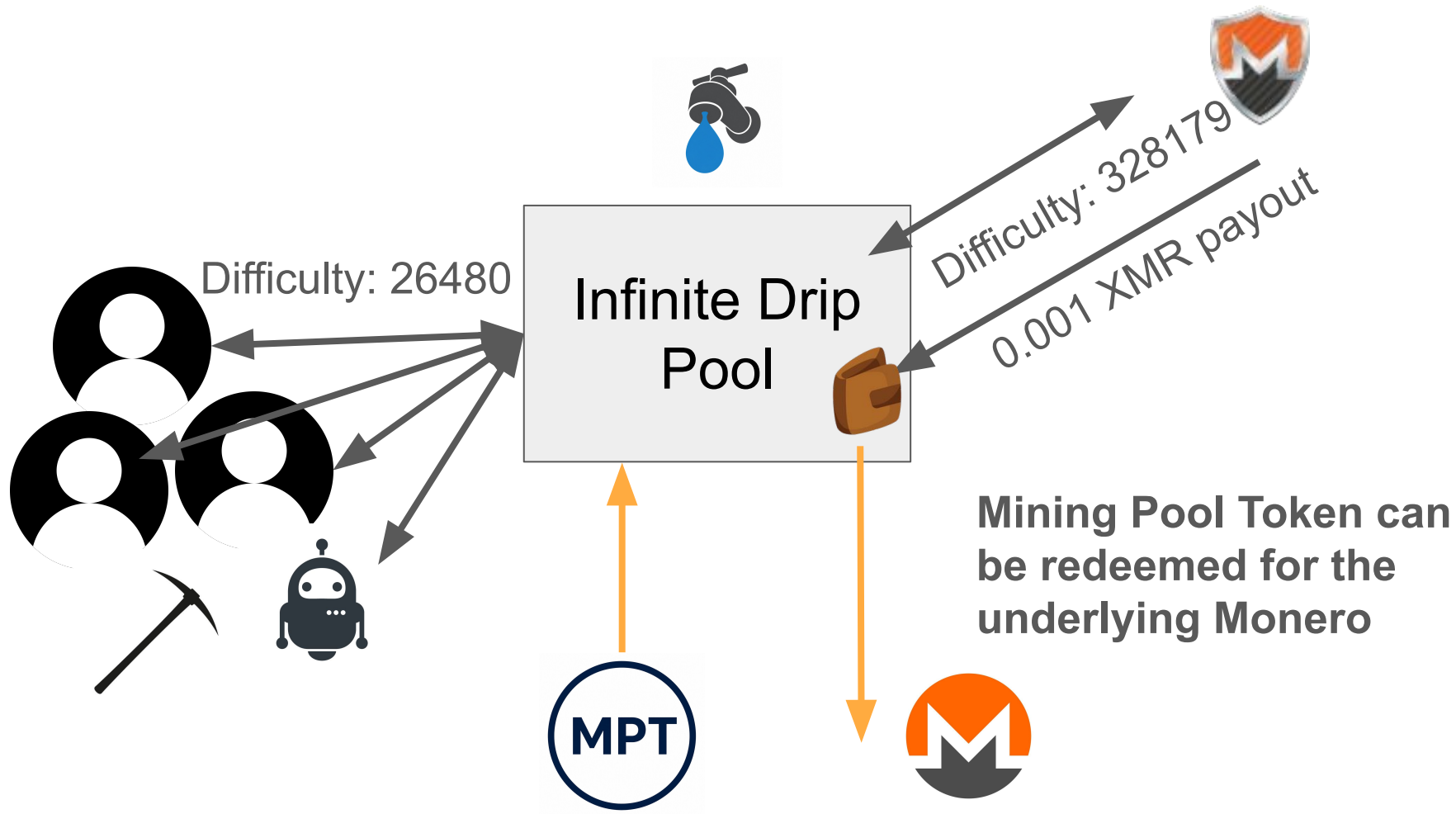


Crossroads

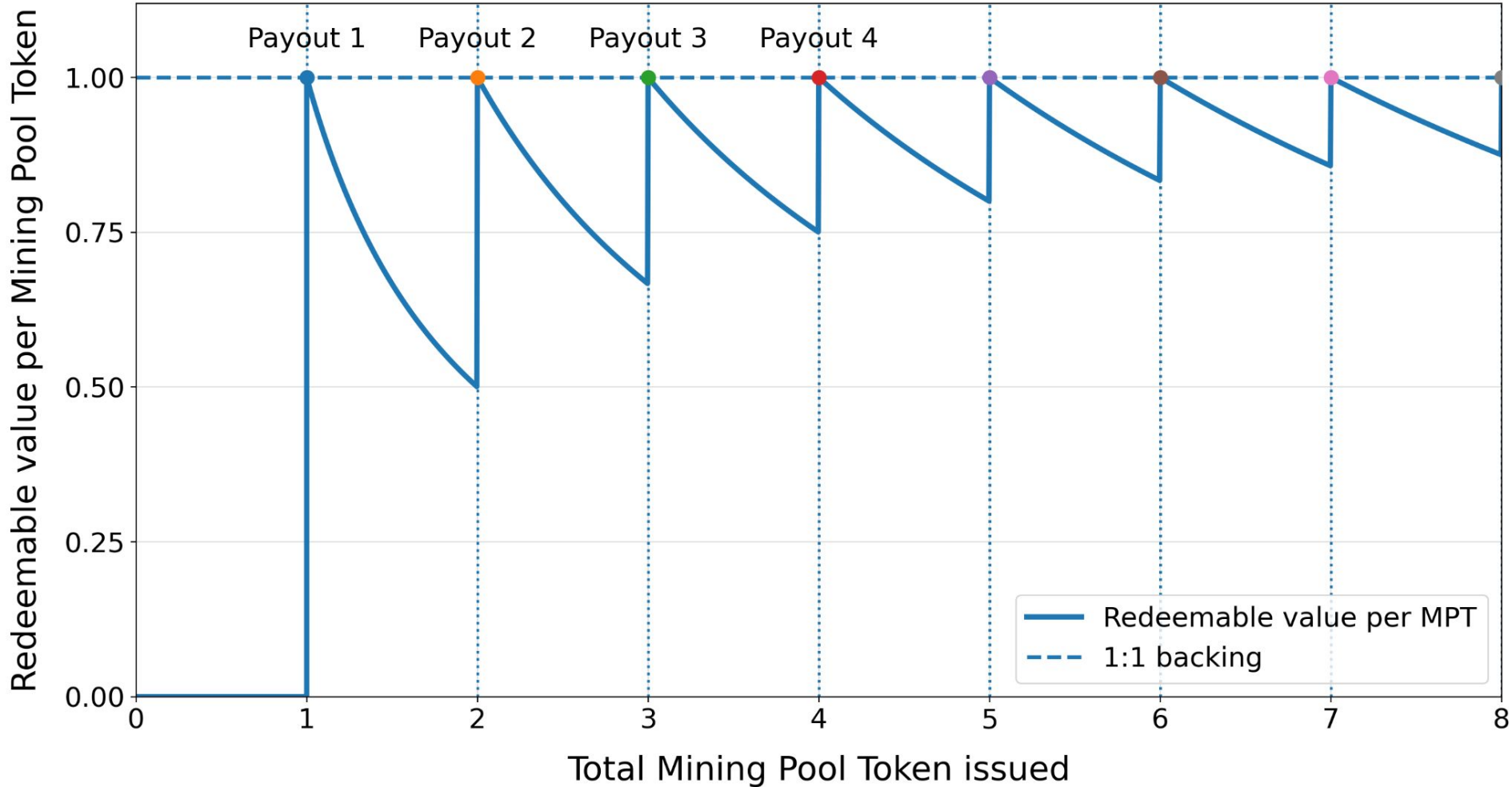
- Lets us represent assets on other chains as ERC-20s
- Integrate with AMMs to trade for them



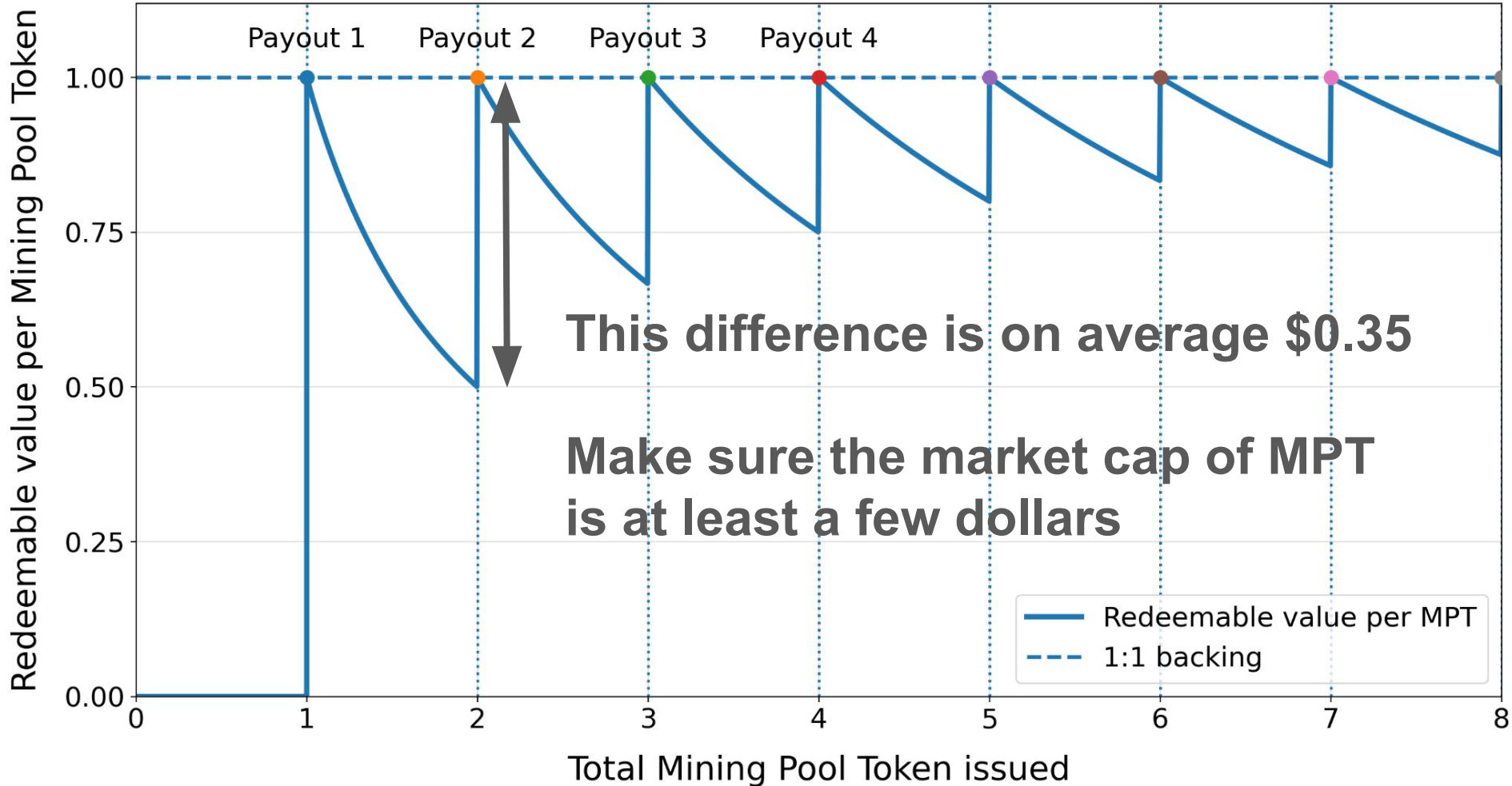




Mining Pool Token Redeemable Value Under Discrete XMR Payouts



Mining Pool Token Redeemable Value Under Discrete XMR Payouts



Entirely self-funding, autonomous program

- Runs in a ROFL container
- Swaps its own fees (in MPT) for ROSE
 - Frontrunning resistance by swapping at random time
- Pool fee:
 - 5% when balance is high
 - Scales up to 40%

[0x46Bd8972BF0348F06De9BA8065E82DC6c15E8006](#) 

Token (ERC-20)

From [0x555F...464a02](#) To  Uniswap V2 For <0.01 MPT

Token (ERC-20)

From  Uniswap V2 To [0x5fd0...FbC2EF](#) For 198.56 WETH

0 TEST

Entirely self-funding, autonomous program

- Cheapest = \$4 / month
- 3GB storage, 2GB RAM, 1 vCPU
- Need 4 weak (laptop) active users on average
 - 1 recent desktop user on average

```
chenghua@Chengs-MacBook-Pro-6 Universal-Faucet % drip start --threads 2
Mining started
address: 0x1a27ad750df161602e7da9280ea4d09351cc4eba
identity: existing
pid: 44680
threads: 2
pool: stratum+ssl://p3333.m269.opf-mainnet-rofl-55.rofl.app:443
voucher: every 300s
helper: pid 44681
log: /Users/chenghua/Library/Application Support/drip/xmrig.log
```

```
Observe:
drip status
chenghua@Chengs-MacBook-Pro-6 Universal-Faucet % drip status
Local miner
status: running
pid: 44680
```

```
Identity
address: 0x1a27ad750df161602e7da9280ea4d09351cc4eba
```

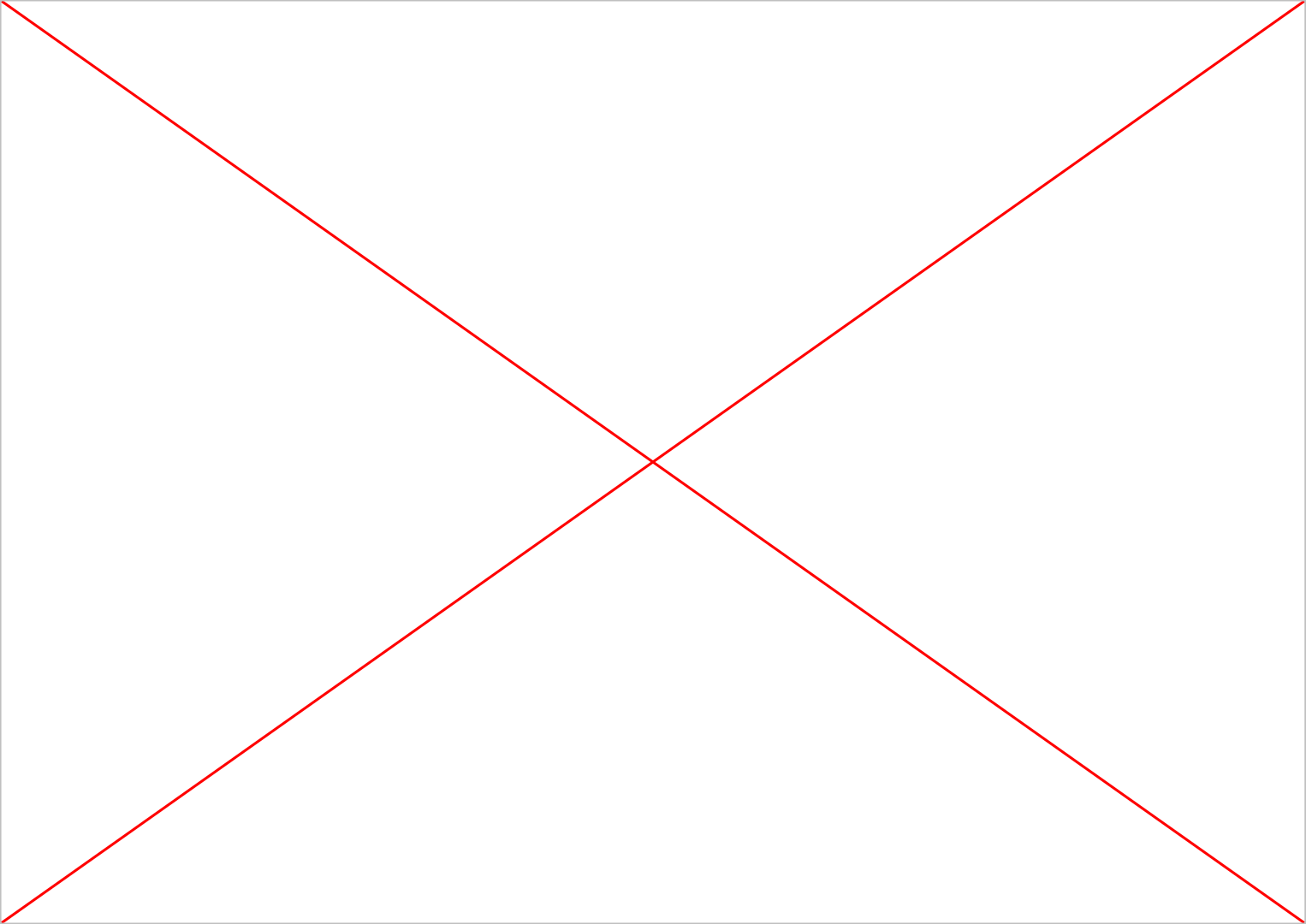
```
Pool
upstream: connected
hashrate: 27,055.25 H/s
active miners: 5
total work: 178,609,467
tor stratum: vj3o34twitcqk7jxopms5mpoxe
```

```
Miner credit
owed atomic xmr: 252,477
voucher watermark: 252,477
shares: 60
work: 1,200,000
last share ms: 1,780,857,667,809
```

```
Voucher
cached cumulative: 252477
signed at: 1780858312
```

CLI Tool

```
Miner credit
owed atomic xmr: 252,477
voucher watermark: 252,477
shares: 60
work: 1,200,000
last share ms: 1,780,857,667,809
```

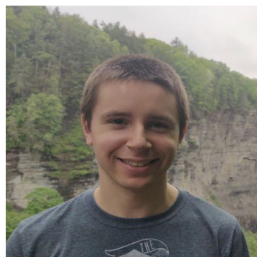


Demo

Future work

- Andrew-level application verifiability
- Tighter DoS protection
- P2P (decentralized) payout relay
- Off-chain proofs to reduce withdrawal cost

Questions



James
Austgen



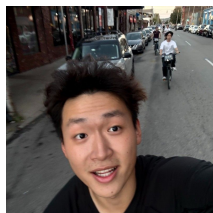
Kaiyang
Chang



Tom
Shangguan



Rithwik
Kerur



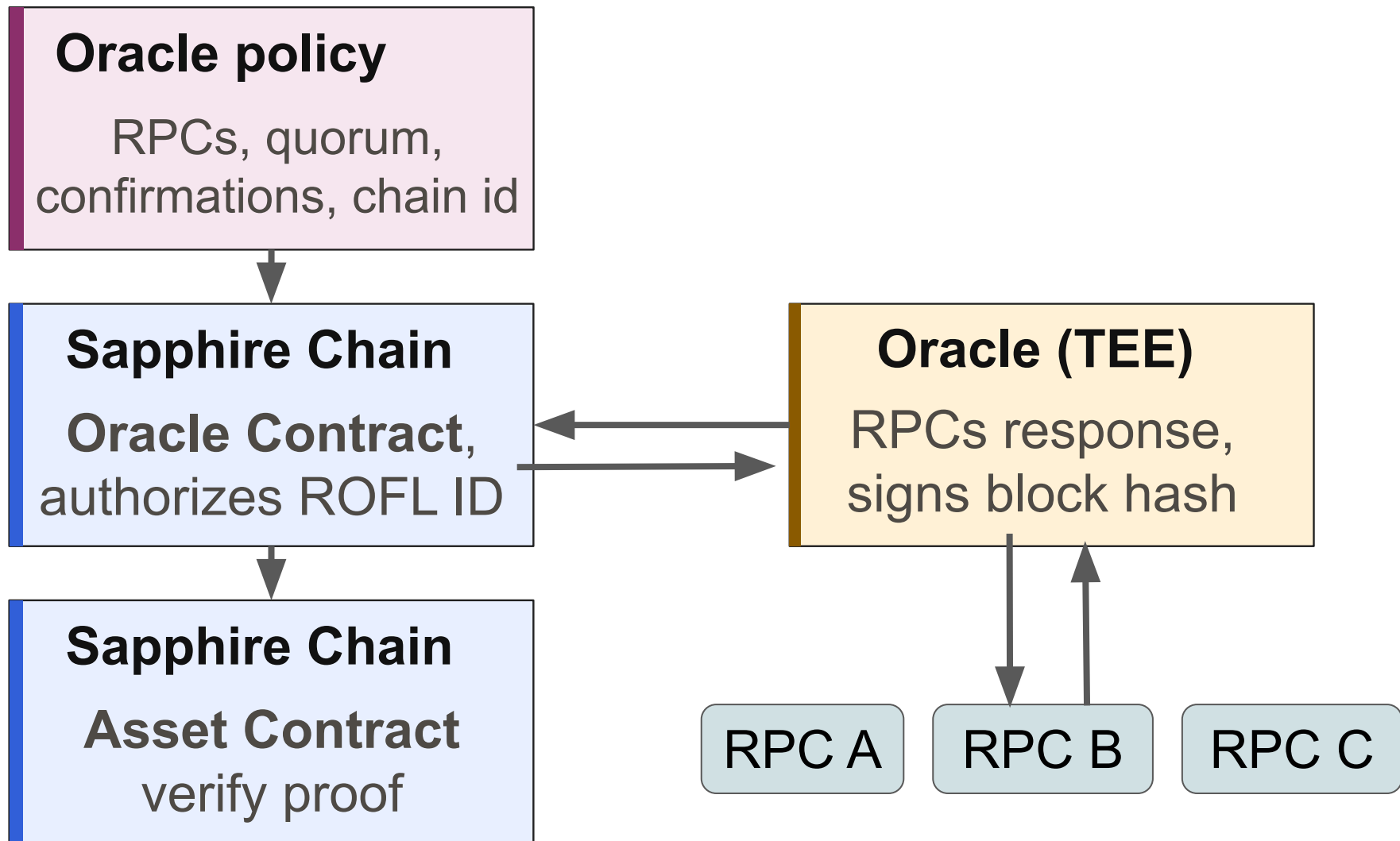
Cheng
Hua



Terry
Huang



Widad
Boulos

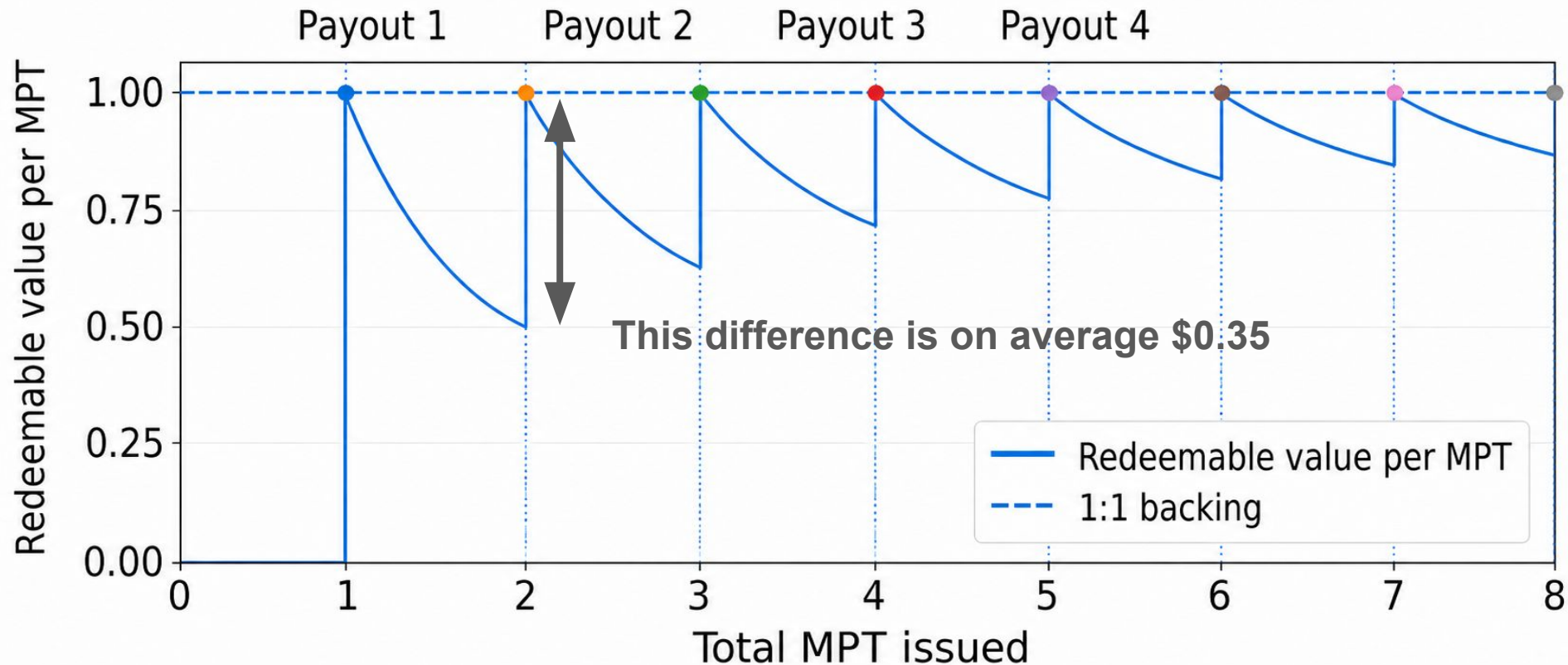


User POV (I think we should highlight how simple this is from the users POV, despite the complicated backend)

- All the user provides is an address on their preferred destination chain.
- They then “mine” until they have how many ever testnet tokens they want

Replace
with demo

MPT Redeemable Value Under Discrete XMR Payouts



Applications

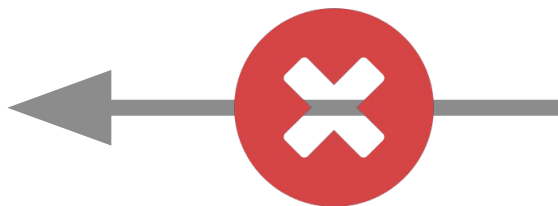
- AI Agents
- Testing smart contracts and blockchain applications
- Micropayments
 - x402
-

Problem



New agent wallet

no gas · no tokens



no way in

**Every funding route
needs a human**



Exchange —
needs KYC



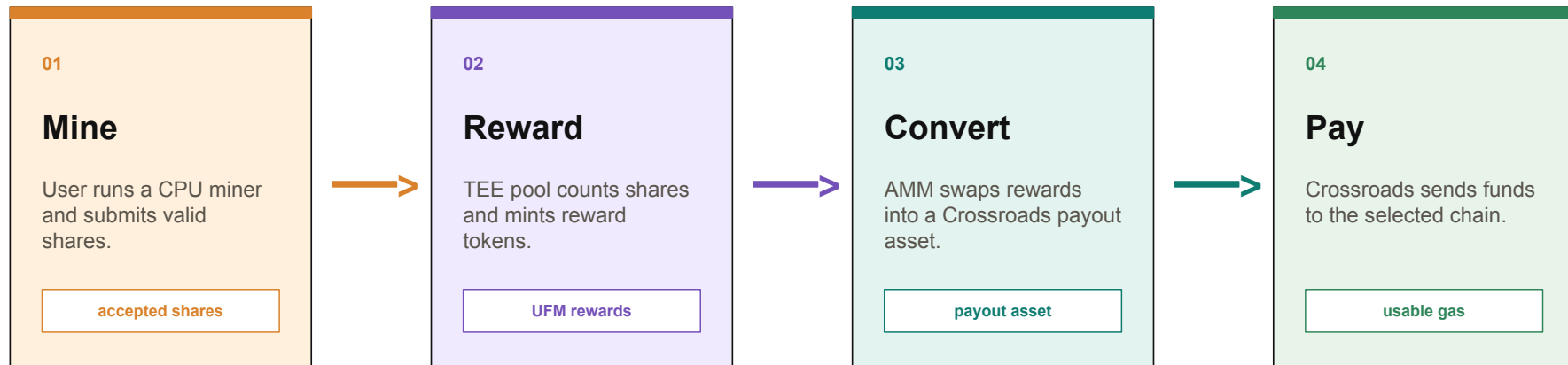
Debit / credit card



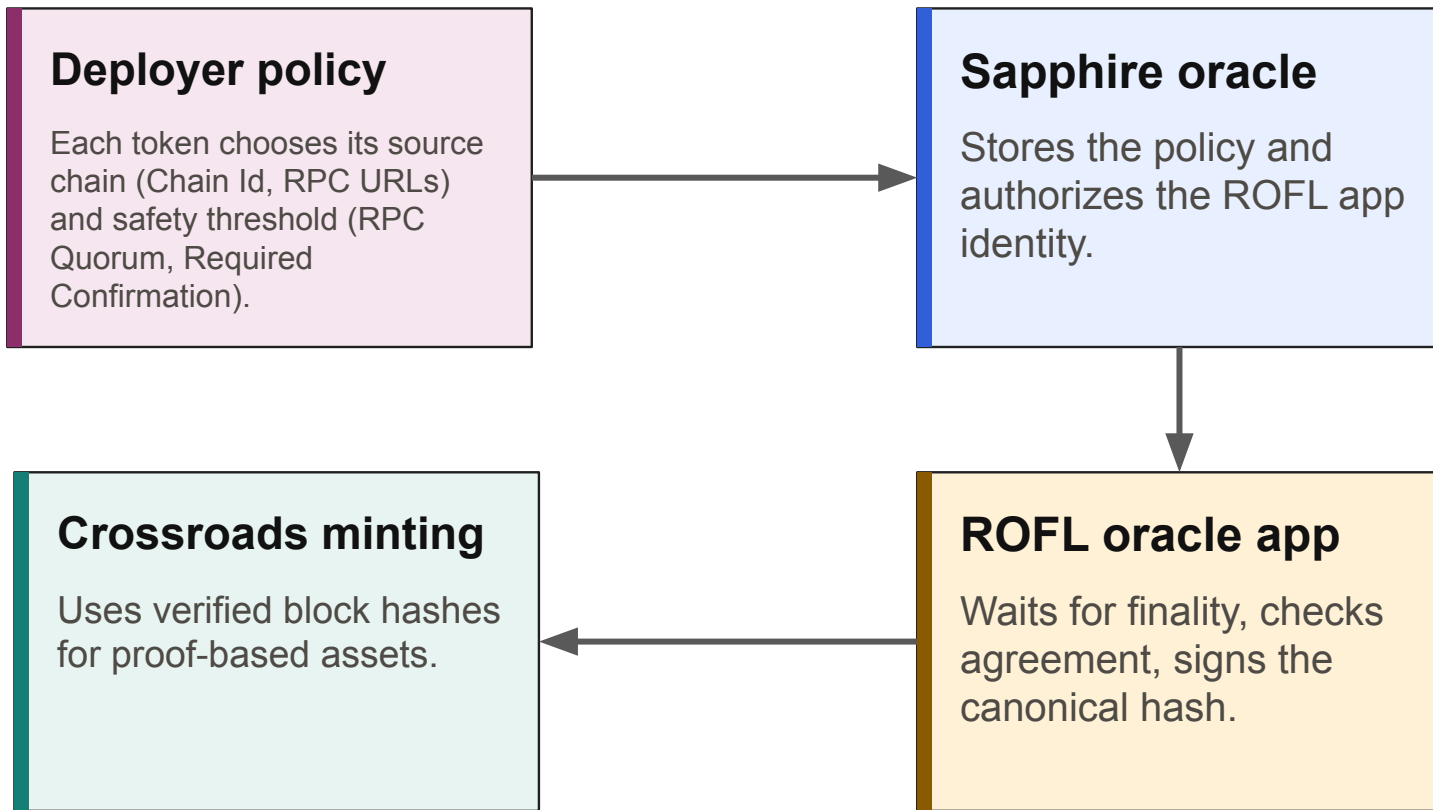
Already hold crypto



A user starts with no funds. The system measures CPU mining work, issues rewards, converts them through Crossroads, and pays gas on the target chain.

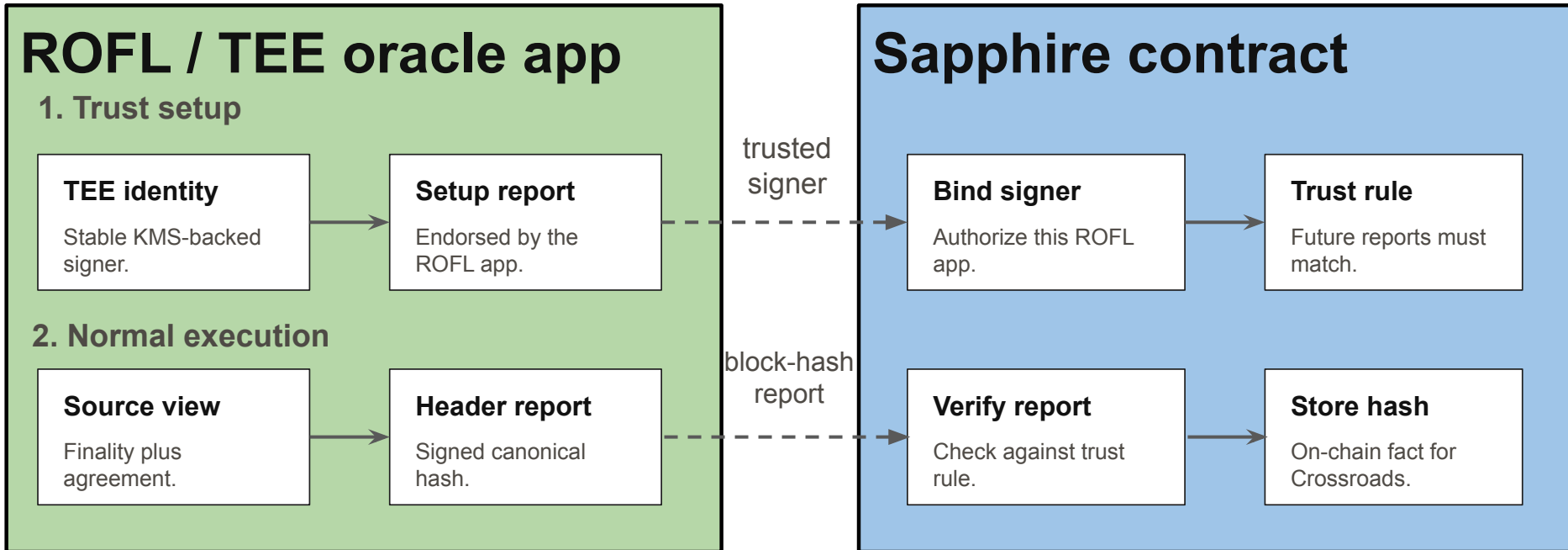


Crossroads: One oracle layer serves many chains



TEE and Sapphire interact in two phases

First the oracle establishes a trusted signer; then normal block-hash reports are verified against that trust binding.



Our Approach (Infinite Drip)

- User does PoW to mine Monero, get MiningPool Token that they can use to swap to the testnet token of their choice
- Benefits are the user doesn't need to do identify verification or own any crypto already

Payout Scheme

- ESMPPS-style deferred PPS
 - With pro-rata settlement
 - Doesn't need large reserve
 - Avoids bankruptcy
- self-funding treasury
- TEE payment bucket
- and TEE-verified accounting

Benefits

- Allows for AI Agents to develop on testnet
- Self regulating pricing via arbitrage bots

